

MATPredict campaign report

Whole-clade runs on the BFD genomes with code v0.6.0 (2026-10-03 to 05). Generated 2026-10-05 from committed tables by results/2026-10-05_campaign_overview/make_dashboard.py.

How to read this report

MATPredict finds mating-type (MAT) loci in fungal genomes. This report summarises three whole-clade runs of code version 0.6.0 over the BFD genome collection (Ascomycota, Basidiomycota and the Mucoromycotina part of Mucoromycota), then looks at two kinds of change: gene order near the MAT region in Xylariales, and locus length and gene content across groups. Everything is drawn from tables committed in the repository; nothing here is new computation.

Called	A genome with at least one MAT call. This measures detection, not truth. Most clades have no curated reference locus, so an uncalled genome is a reference gap until shown otherwise.
Routed by lineage / phylum fallback	Genomes whose order has its own curated record are searched with it (blue). Others are searched against the whole phylum (orange) and call less reliably.
Locus class	Full locus: the MAT gene(s) plus the expected flank structure. Partial locus: some of it. Gene only: the idiomorph gene without a recognised locus. Homothallic candidate: both idiomorphs at one locus.
Confidence	High, medium or low, from how complete the evidence is.
PR-only calls	Basidiomycota genomes whose only call is the pheromone-receptor family. 1,360 of these come from a motif scan alone and are labelled unverified.
SAC and SCA (Xylariales)	Orders of the flank genes SLA2, COX13 and APN2. SAC keeps the outgroup order with an open interval between SLA2 and APN2 where a MAT locus sits. SCA has COX13 between SLA2 and APN2, so the interval is closed.

Panel by panel

- **Summary.** Headline counts; call rate per class or order (a marker shows the rate without PR-only calls); and the composition of the calls. Takeaway: rates are high where a clade has its own record and low where it does not.
- **Exemplars and outliers.** Short cards with a source file each. Exemplars show what the approach recovers; outliers show where it falls short or where biology differs.
- **Xylariales synteny.** A gene-order map of seven genomes on one frame; the sampling bar chart (genomes versus species); a table of which neighbours remain adjacent; and a histogram of the SLA2 to APN2 distance. Takeaway: the NC1011 order is shared by many species and its MAT interval is closed.
- **Locus size and content.** Locus length per Mucoromycotina genus and per filamentous Ascomycota class (bar = interquartile range, dot = median, line = range), a table of which genes each Mucoromycotina genus carries, and the share of Ascomycota loci carrying both flank genes.

Limits: counts are descriptive; several runs used earlier code; genome counts reflect uneven strain sampling; synteny is gene-level only. The markdown overview is analysis/2026-10-05_campaign-overview.md and the guide is analysis/2026-10-05_campaign-dashboard-guide.md.

1. Summary

GENOMES SEARCHED

22,974Ascomycota 19,415 · Basidiomycota
3,270 · Mucoromycotina 289

MAT LOCI REPORTED

27,176

21,011 · 5,900 · 265

GENOMES WITH A CALL

89% · 91% · 89%Ascomycota · Basidiomycota ·
Mucoromycotina; detection, not
ground truth

HIGH-CONFIDENCE LOCI

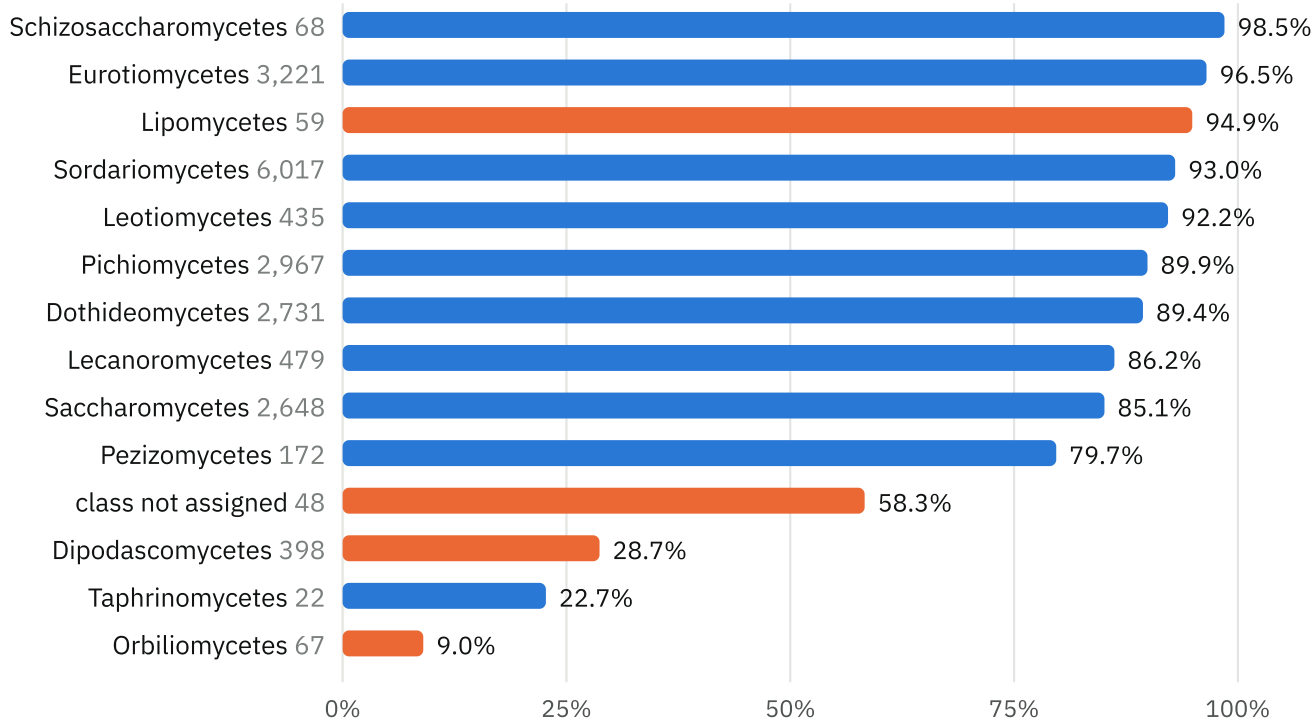
65% · 65% · 77%

same order of campaigns

Call rate by Ascomycota class

Classes with 20 or more genomes (19,332 of 19,415). Orange classes had no curated record for their clade and were searched against the whole phylum, so a low rate there is a reference gap, not a measured absence.

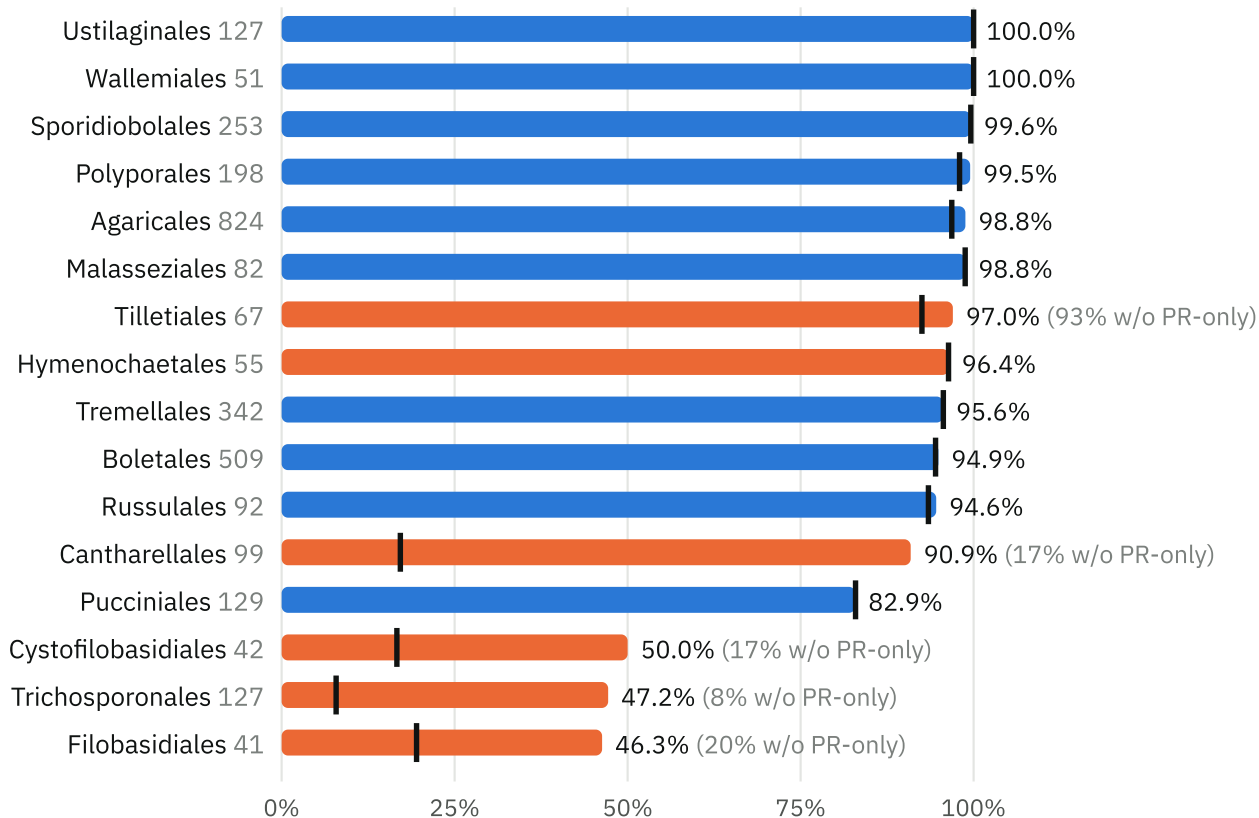
■ routed by lineage ■ mostly phylum fallback



Call rate by Basidiomycota order

Orders with 40 or more genomes. The marker is the rate after removing PR-only calls; 1,360 PR calls come only from a CAAX-motif scan and are labelled unverified. Cantharellales falls from 91% to 17% and Trichosporonales from 47% to 8%.

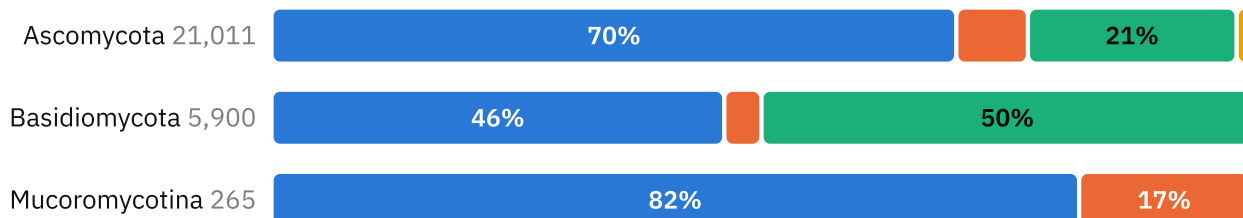
■ routed by lineage ■ mostly phylum fallback



What the calls contain

Share of loci by class. Half of the Basidiomycota loci are the idiomorph gene alone, without a recognised locus around it.

■ full locus ■ partial locus ■ gene only ■ homothallic candidate



2. Exemplars and outliers

EXEMPLAR

Wallemiales: 0 to 51 of 51

No MAT record existed. One record built from an undeposited locus calls every one of the 51 genomes (33 high, 18 medium) and separates two versions of the locus; the four *W. ichthyophaga* genomes in the minority are the published inverted strains.

`docs/notes/publication-highlights.md`

EXEMPLAR

Rhodotorula P/R: 61 of 62 held-out strains

The called P/R allele matches an independent study for 61 of 62 held-out strains; the miss is a reported hybrid. HD calls rose from 0 to 217 genomes.

`docs/publication-notable-findings/002`

EXEMPLAR

Lineage-routed orders at 95 to 100%

Ustilaginales 100%, Wallemiales 100%, Sporidiobolales 99.6%, Polyporales 99.5%, Agaricales 98.8%: orders with their own curated record.

`results/2026-10-03_basidiomycota_v060/by_order.tsv`

EXEMPLAR

Zygomycete benchmark 23 of 23

All 23 Zygo genomes are typed on both scaffold and contig input, and the regression panel shows no changed locus across five panels.

`results/2026-10-03_regression_v060`

OUTLIER

Reference gaps: Orbiliomycetes 9%, Dipodascomycetes 29%

Both classes are searched through the phylum fallback. Low rates mark missing curation.

`results/2026-10-03_ascomycota_v060/by_class.tsv`

OUTLIER

PR-only calls inflate fallback orders

Cantharellales 91% drops to 17% and Trichosporonales 47% to 8% without PR-only calls. The CAAX-only calls are under a chance-level test in a draft PR (#33): about 13% of the unverified calls sit at chance.

`analysis/2026-10-05_caax-receptor-test.md (draft)`

OUTLIER

Dothideomycetes: SLA2 detached (6% with both flanks)

Only 6% of full Dothideomycete loci carry both APN2 and SLA2, against 82 to 91% in the other filamentous classes. A genome-wide search finds SLA2 in every sampled genome but inside the locus in only 14% (80% in Sordariomycetes); Cladosporiales are the exception. Mostly organisation, not missed detection.

`analysis/2026-10-05_dothideomycetes-sla2.md`

OUTLIER

Early-diverging fungi: few calls

Mortierellomycota 6 of 100, Kickxellomycota 7 of 190, and Lichtheimiaceae 1 of 30 in the 2026-09-26 scan. No Mucorales-type synteny support in *Mortierella* or *Kickxella*.

`results/2026-09-26_early_diverging/summary.txt`

OUTLIER**Xylaria NC1011: flanks, no MAT gene**

SLA2, COX13 and APN2 are intact and expressed, with 475 and 118 bp between them and no HMG-box or alpha-box domain in the 47-kb region. The one call is an HMG gene nearest NCU03481 (14% support).

`analysis/2026-10-05_xylariales-syteny.md`

OUTLIER**Mycotypha: sexM 153 kb from sexP**

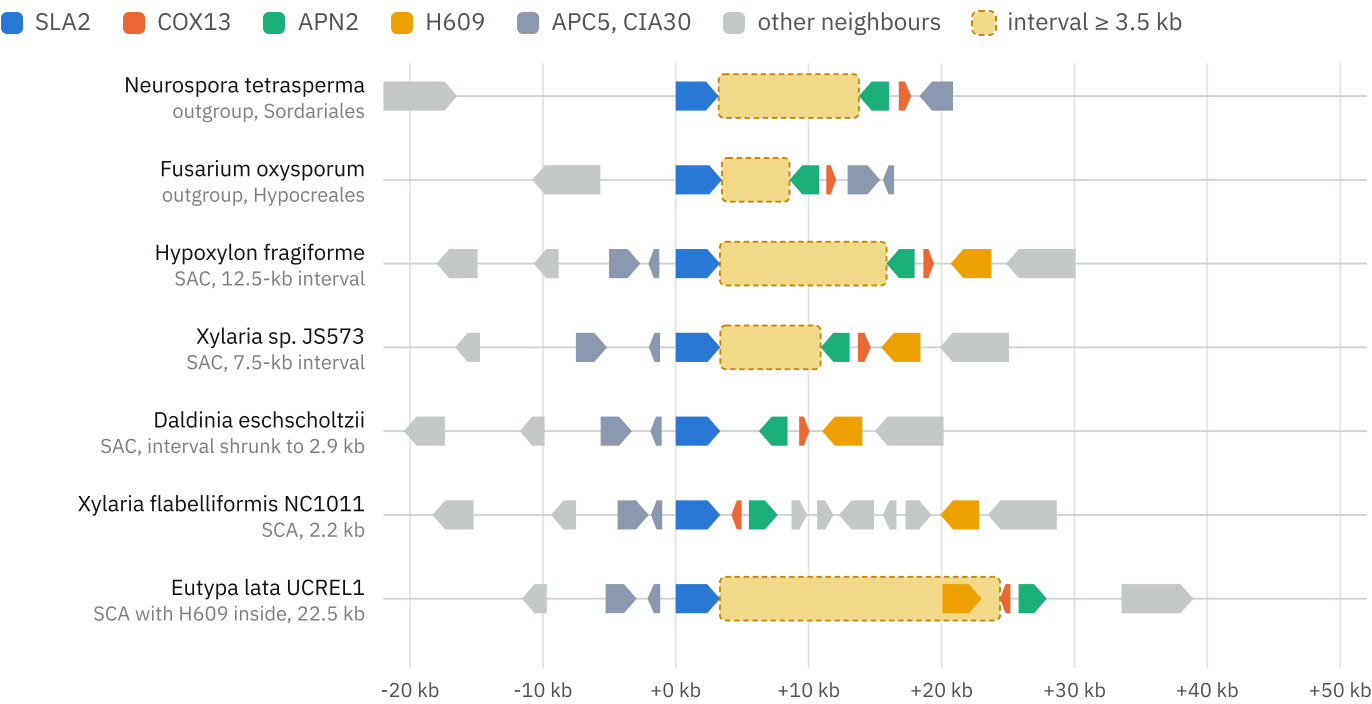
A homothallic species with its two idiomorph genes about 150 kb apart on one scaffold. v0.6.0 did not call the sexM gene and the classifier carried it among its negatives; the classifier was rebuilt on 2026-10-04 and the new outcome has not been re-checked here.

`analysis/2026-10-04_unclassified-hmg-in-gene-trees.md`

3. Xylariales synten

Gene order near the MAT region in Xylariales

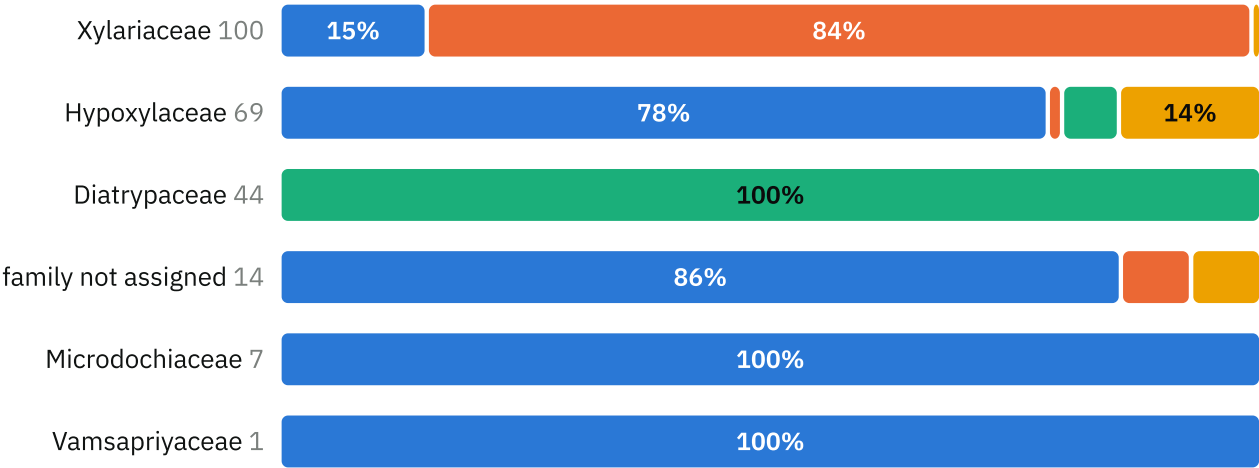
Seven genomes drawn on the same frame, SLA2 at zero and pointing right. Gold boxes mark the interval between SLA2 and the next flank gene where a MAT locus sits in outgroups. In outgroups and the SAC genomes the interval is open; in NC1011 it holds only COX13.



Sampling: 257 genomes, 164 species

41 of the 257 genomes are strains of *Eutypa lata*, all in the SCA state with H609 inside the interval, so genome counts overstate how often SCA arose. By species, SAC is 69 and SCA is 73 (majority state per species; 9 other, 13 incomplete).

■ outgroup-like order (MAT interval kept) ■ COX13 beside SLA2 ■ COX13 beside SLA2, plus H609 inside
■ other order



Which neighbours stay together

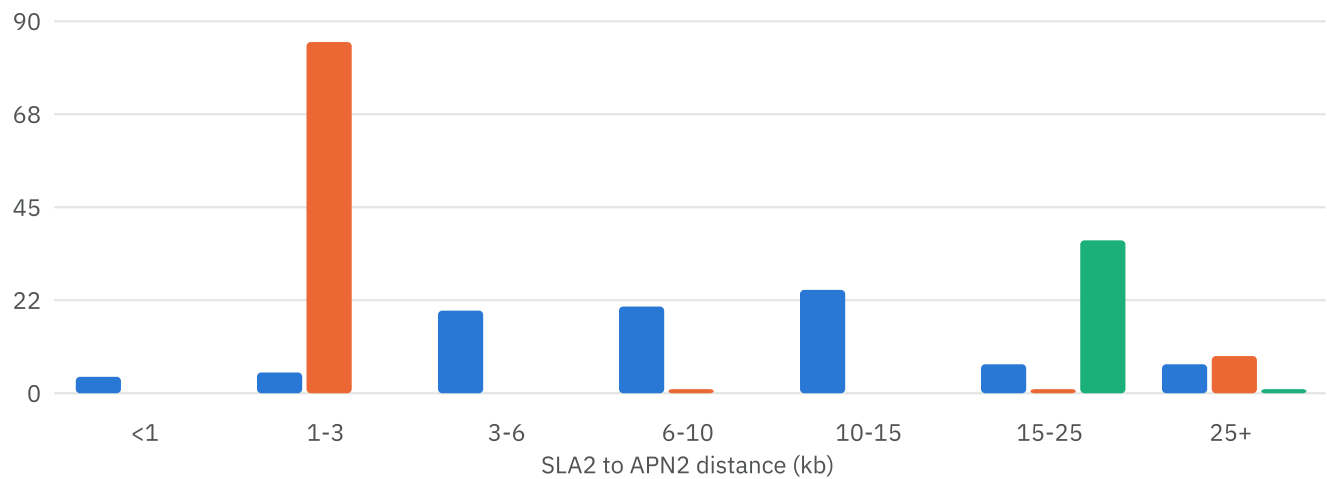
Share of genomes in which two NC1011 neighbours are still adjacent with the same relative orientation. Two junctions change: CIA30 next to SLA2 (absent in outgroups, shared with Amphisphaeriales), and SLA2 next to COX13 (only in SCA). COX13 and APN2 never separate.

Neighbours in NC1011	outgroups	Amphisphaeriales	Xylariales SAC	Xylariales SCA
ARM-FRE	100%	80%	99%	100%
FRE-APC5	0%	1%	99%	100%
APC5-CIA30	39%	100%	99%	99%
CIA30-SLA2	0%	100%	100%	100%
SLA2-COX13	4%	0%	0%	64%
COX13-APN2	100%	100%	100%	100%

How much room is there between SLA2 and APN2

SAC keeps a gap of several kb, enough for a MAT locus; SCA genomes mostly have only COX13 between them. The deletions in a few SAC genomes (about 3 kb) are a second route to losing the region.

■ SAC, n=89 ■ SCA excluding E. lata, n=96 ■ Eutypa lata, n=38

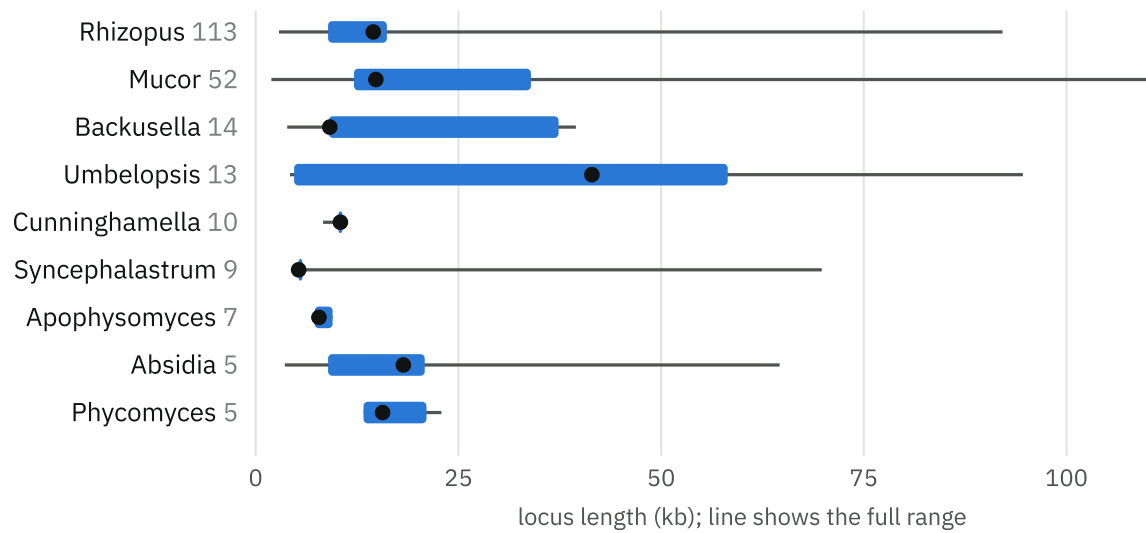


Sources: results/2026-10-05_xylariales_nc1011_interval, analysis/2026-10-05_xylariales-syntenymd.
Gene-level only; breakpoints are bounded by gene ends.

4. Locus size and content

Mucoromycotina: locus length by genus

Genera with five or more called loci in the BFD set. Bar is the interquartile range, dot the median, line the full range. Median across all 265 loci is 14.5 kb. *Umbelopsis* has the longest loci (median 41.5 kb); *Syncephalastrum* and *Apophysomyces* have the shortest (5.3 and 7.8 kb).



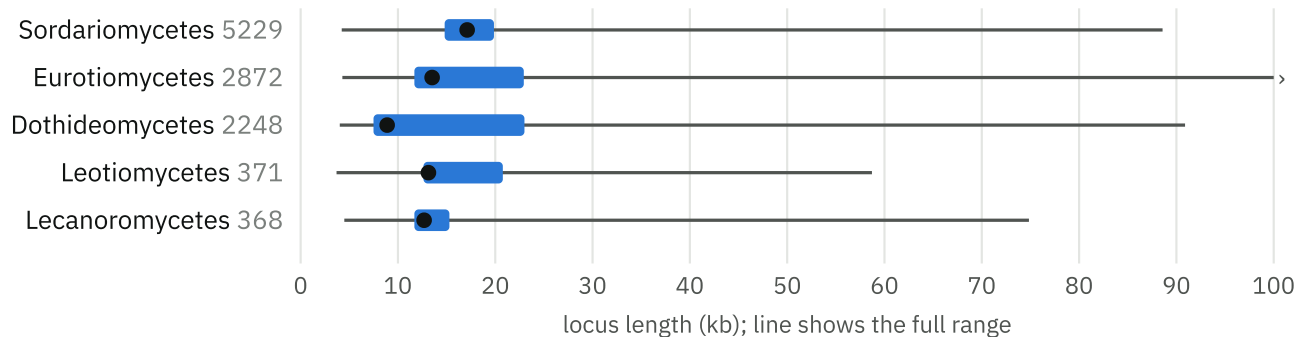
Mucoromycotina: which genes the locus carries

Share of each genus's loci in which the gene was found. Content differs by genus: *btbA* is found only in *Rhizopus* (77% of its loci); *glrA* is missing from *Cunninghamella*, *Syncephalastrum*, *Apophysomyces* and *Phycomyces*; *rnhA* is missing from *Backusella* and *Apophysomyces*; *tptA* is missing from *Syncephalastrum*. Percentages for *sexP* and *sexM* can add to more than 100 because a locus can list both.

Genus (loci)	sexP	sexM	tptA	rnhA	algA	glrA	btbA
Rhizopus 113	59%	68%	51%	83%	4%	76%	77%
Mucor 52	48%	58%	85%	96%	85%	85%	0%
Backusella 14	71%	29%	29%	0%	100%	93%	0%
Umbelopsis 13	69%	54%	100%	46%	100%	54%	0%
Cunninghamella 10	70%	30%	100%	100%	90%	0%	0%
Syncephalastrum 9	78%	22%	0%	100%	11%	0%	0%
Apophysomyces 7	100%	29%	100%	0%	100%	0%	0%
Absidia 5	20%	80%	40%	20%	80%	60%	0%
Phycomyces 5	40%	60%	100%	100%	100%	0%	0%

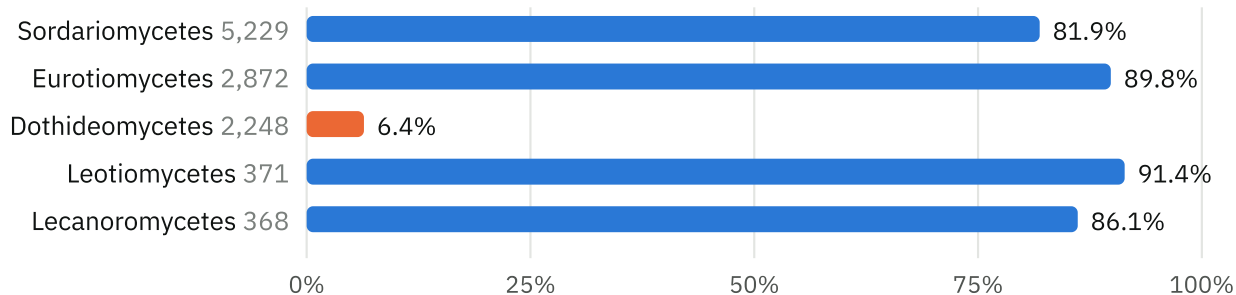
Filamentous Ascomycota: locus length by class

Full loci of the generic Ascomycota MAT family. Sordariomycetes loci are twice as long as Dothideomycete loci at the median (17.1 against 8.9 kb), and Dothideomycetes have a long upper tail.



Filamentous Ascomycota: APN2 and SLA2 in the same locus

Share of full loci that carry both flank genes. Dothideomycetes stand out.

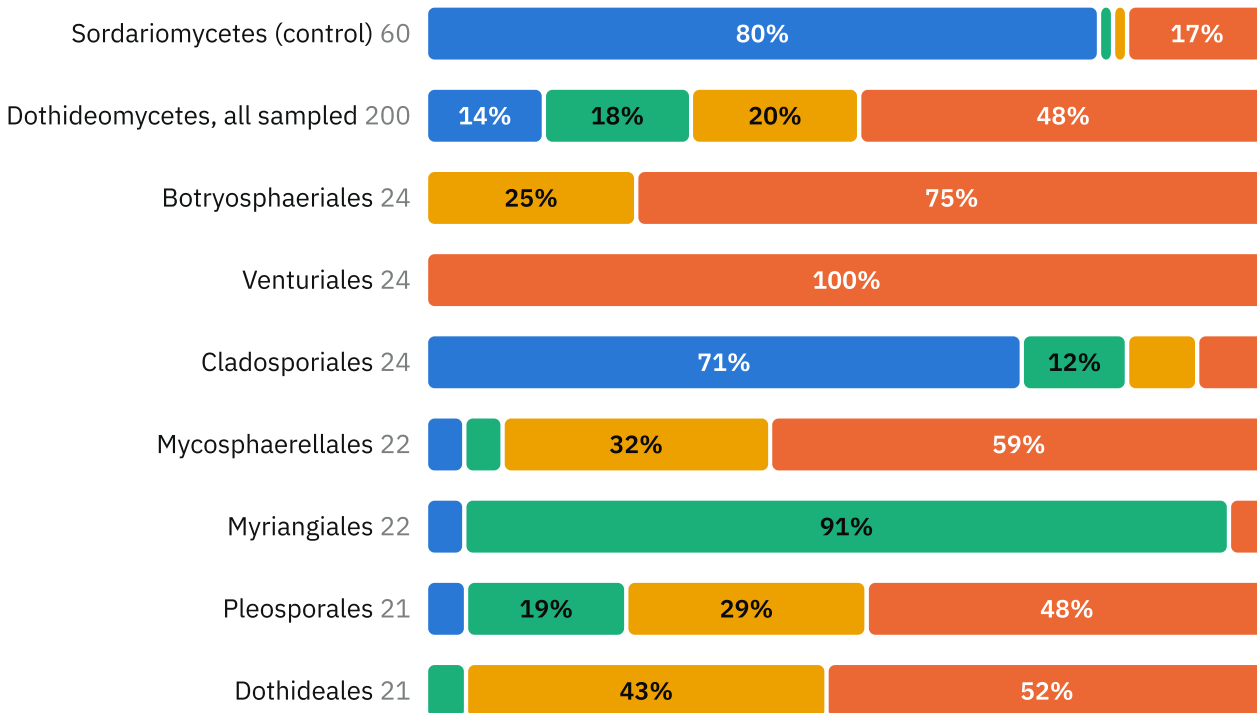


Sources: results/2026-10-03_mucoromycotina_mat/calls.tsv (BFD, called), results/2026-10-03_ascomycota_v060/loci.tsv.

Dothideomycetes: SLA2 is detached from the MAT locus, not missed

A genome-wide search for SLA2 in 200 sampled Dothideomycete genomes finds it in 200 (best hit 85% positives against the NC1011 protein), so the gene is present and recognisable. Only a minority place it inside the called locus, against 80% in the Sordariomycete control. Cladosporiales keep it there; in most other orders SLA2 sits tens of kb away or on another contig. "Another contig" includes assembly breaks and misses (17% in the control). Sample: up to 25 genomes per order, one locus each.

■ inside the called locus ■ 20 to 100 kb away, same contig ■ over 100 kb away, same contig ■ on another contig



Source: results/2026-10-05_dothideo_sla2_test/sla2_distance.tsv (miniprot, SLA2 query from Xylaria NC1011), analysis/2026-10-05_dothideomycetes-sla2.md.